Experiment 1

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A. Set up Linux - simple guide

The easiest way to use Linux without changing Windows is to run it inside a virtual machine. VirtualBox is the app that opens the Linux computer; Kali Linux is the Linux system inside it.

1. Download VirtualBox

Go to the official VirtualBox download page and click Windows hosts. Save the installer.

2. Install VirtualBox

Open the downloaded installer. Keep the normal settings, click Next until it finishes, then open VirtualBox.

3. Download Kali Linux

Go to the official Kali Linux page. Choose Virtual Machines, then select the VirtualBox download.

4. Extract the Kali file

The Kali download is usually a ZIP file. Right-click it and choose Extract All. Keep the extracted files in the same folder.

Open Kali Linux in VirtualBox

5. Add the Kali machine

In VirtualBox, click Machine > Add. Open the extracted folder and select the file ending in .vbox. The Kali machine will appear on the left.

6. Start Kali

Click the Kali machine once, then click the green Start button. Wait for the Kali login screen and sign in using the details supplied with the download or by your teacher.

7. Open a terminal

After the desktop appears, open Terminal. This is where you type Linux commands.

Check that Linux is ready:

```
whoami
pwd
```

The first command shows your username. The second command shows the folder you are currently using.

Small safety tip: Create a snapshot in VirtualBox before trying new software. It gives you a restore point if something goes wrong.

Official download pages: https://www.virtualbox.org/wiki/Downloads/ | https://www.kali.org/get-kali/

B. Working with Linux Commands - Worked Solution

The examples below assume a fresh practice folder. Type each command after the $ prompt. Output can vary slightly by computer, username, and date.

First create a safe practice area:

```
$ mkdir -p ~/linux_lab
$ cd ~/linux_lab
$ printf "alpha\nbeta\ngamma\n" > notes.txt
```

Terminal basics

pwd - Prints the full path of the current working directory.

```
$ pwd
/home/kali/linux_lab
```

mkdir - Creates a new directory named projects.

```
$ mkdir projects
```

rmdir - Removes an empty directory. It fails if the directory contains files.

```
$ rmdir projects
```

dir - Lists directory contents in a basic format; it is similar to ls.

```
$ dir
notes.txt
```

ls - Lists files and directories. Try ls -la to include hidden files and details.

```
$ ls
notes.txt
```

cd - Changes to the specified directory.

```
$ cd /tmp
$ pwd
/tmp
```

cd .. - Moves up one level to the parent directory.

```
$ cd ..
$ pwd
/home/kali
```

clear - Clears the visible terminal screen; it does not delete command history.

```
$ clear
```

history - Shows previously entered shell commands.

```
$ history | tail -3
12 ls
13 clear
14 history | tail -3
```

man - Opens the manual page for ls. Press q to quit.

```
$ man ls
```

whoami - Displays the name of the currently logged-in user.

```
$ whoami
kali
```

Files and permissions

vi - Opens the vi text editor. Press i to insert, Esc then :wq to save and quit.

```
$ vi notes.txt
```

file - Identifies a file type from its contents.

```
$ file notes.txt
notes.txt: ASCII text
```

rm - Deletes a file. Be careful: rm normally does not use a recycle bin.

```
$ rm copy.txt
```

cp - Copies notes.txt to a new file named copy.txt.

```
$ cp notes.txt copy.txt
```

mv - Renames a file, or moves it when the destination is another directory.

```
$ mv copy.txt backup.txt
```

cat - Prints a file’s contents to the terminal.

```
$ cat notes.txt
alpha
beta
gamma
```

more - Views a text file one screen at a time; press q to exit.

```
$ more notes.txt
```

less - Views a file with forward/backward scrolling and search; press q to exit.

```
$ less notes.txt
```

head - Displays the first lines of a file; default is 10 lines.

```
$ head -2 notes.txt
alpha
beta
```

tail - Displays the last lines of a file; default is 10 lines.

```
$ tail -2 notes.txt
beta
gamma
```

chmod - Changes permissions. u+x grants execute permission to the file owner.

```
$ chmod u+x script.sh
$ ls -l script.sh
-rwxr--r-- 1 kali kali 0 Aug 19 script.sh
```

Redirection

> - Redirects output into a file, creating it or replacing its previous contents.

```
$ echo "first line" > output.txt
```

>> - Appends output to the end of a file without removing existing contents.

```
$ echo "second line" >> output.txt
$ cat output.txt
first line
second line
```

< - Sends a file’s contents to a command as standard input. Here, wc counts lines.

```
$ wc -l < notes.txt
3
```

Clean up the practice files (optional)

```
$ cd ~/linux_lab
$ rm -f notes.txt backup.txt output.txt script.sh
$ cd ..
$ rmdir linux_lab
```

This removes only the files created for the exercise. Check the folder name carefully before running rm or rmdir.